

Two Methods for Scope Management

1. Scope Baseline Method

The scope baseline method defines the project's boundaries at the start — what's included and what's not. It includes three main documents: the project scope statement, the Work Breakdown Structure (WBS), and the WBS dictionary. Together, these form a “baseline” that can only be changed through a formal approval process.

Pros:

- Prevents uncontrolled changes (scope creep).
- Keeps all stakeholders aligned on what the project actually includes.
- Makes it easier to measure progress against the original plan.

Cons:

- Can be too rigid if changes are needed quickly.
- Requires formal documentation and approvals, which can slow down the project.

Pseudocode Example:

```
function manageScopeBaseline(requestedChange):  
    if requestedChange in baseline:  
        print("Change already part of baseline.")  
    else:  
        if approveChange(requestedChange):  
            updateBaseline(requestedChange)  
            print("Scope baseline updated.")  
        else:  
            print("Change rejected. Scope remains the same.")
```

```
function approveChange(change):  
    # Simulates a change control board decision  
    return change.importance >= 5 and change.costImpact < 10000
```

2. Agile Scope Management Method

In agile projects, scope is managed dynamically through iterative development. Instead of locking down the full scope upfront, teams maintain a prioritized backlog that evolves over time. Changes are expected and handled through sprint planning.

Pros:

- Very flexible; adapts to changing needs or feedback.
- Encourages collaboration and continuous improvement.
- Allows for quicker delivery of working features.

Cons:

- Can make budgeting and timelines harder to predict.
- Without discipline, changing priorities can lead to confusion.

Pseudocode Example:

```
function manageAgileScope(productBacklog, sprintGoals):  
    for sprint in sprints:  
        sprintScope = prioritize(productBacklog, sprintGoals)  
        executeSprint(sprintScope)  
        feedback = collectFeedback()  
        updateBacklog(productBacklog, feedback)  
    print("Project completed with adaptive scope management.")
```

```
function prioritize(backlog, goals):  
    return [item for item in backlog if item.priority in goals.topPriorities]
```

Summary:

The scope baseline method is best for projects that require strict control and documentation, like government or construction work. The agile method works better for software projects where requirements can change frequently. Each method balances flexibility and control differently depending on the project type.

<https://www.geeksforgeeks.org/project-mgmt/what-is-scope-in-project-management/>